Data Analytics

Sci-Kitchen Luxury Foods: A Predictive Segmentation Study

Claire Leyden Nalpas

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Introduction

This predictive segmentation project aimed to improve the efficiency of Sci-Kitchen Luxury Foods marketing campaigns by identifying customers who have a high customer lifetime value and also those that are most likely to respond to promotional activities. By combining customer demographic, relationship (CRM), and behavioural data with machine learning techniques, the project next sought to identify the customer characteristics that predict marketing conversion. The overall goal was to enable more targeted campaigns, reduce marketing expenditure on low-probability customers, and support improved customer lifetime value and campaign return on investment (ROI).

To achieve this goal, the project had the following objectives: identify the customer characteristics associated with marketing conversion; compare the predictive performance of models built using demographic, relationship, behavioural, and combined feature groups; evaluate whether the modelling methodology produces consistent results in a different business context using a supplementary external dataset; and develop deployable assets including a relational database, API, and Streamlit application.

The project was developed in the following stages:
Data collection: Retrieval of the Sci-Kitchen and UCI Bank Marketing datasets, together with external enrichment data.
Data preparation and EDA: Data cleaning, feature engineering, and exploratory analysis to identify relationships with campaign conversion.
Customer segmentation: K-Means clustering to identify customer segments within the Sci-Kitchen dataset.
Predictive modelling: Development and comparison of logistic regression, random forest and XGBoost models using demographic, behavioural, relationship, and combined feature groups.
Model validation: Statistical comparison of model performance using DeLong tests.
Database and API development: Design and implementation of a MySQL relational database, Entity Relationship Diagram (ERD), SQL queries, and a Flask API to expose the data.
Application deployment: Deployment of a Streamlit application for end-user interaction.
External validation: Application of the modelling methodology to the UCI Bank Marketing dataset to assess the generalisability of the findings.

Data and data sources

Primary Dataset: Customer Personality Analysis

The primary dataset used throughout the project is the Customer Personality Analysis dataset, adapted to represent the Sci-Kitchen Luxury Foods business case. The dataset contains customer-level records describing demographic characteristics, customer relationship history, and purchasing behaviour, making it suitable for customer segmentation and conversion prediction.

The dataset includes:

Demographic data: age, education, marital status, income, and household composition.
Relationship (CRM) data: previous campaign responses, recency of customer interactions, customer complaints, and customer tenure indicators.
Behavioural and transactional data: spending across product categories, purchase channels, purchase frequency, and website activity.

This dataset forms the foundation for feature engineering, customer segmentation, and predictive modelling.

Supplementary Validation Dataset

The UCI Bank Marketing dataset was incorporated as a supplementary validation dataset to evaluate whether the modelling methodology developed for Sci-Kitchen could be applied to a different industry. The dataset contains customer information collected during direct marketing campaigns conducted by a Portuguese banking institution, including demographic characteristics, campaign history, and financial information.

Unlike the Sci-Kitchen dataset, the Bank Marketing dataset contains limited purchasing or behavioural information and therefore is not suitable for behavioural customer segmentation. Instead, it provides an independent case study for validating the predictive modelling approach and comparing the relative contribution of different feature groups within an alternative business environment.

External Data Enrichment

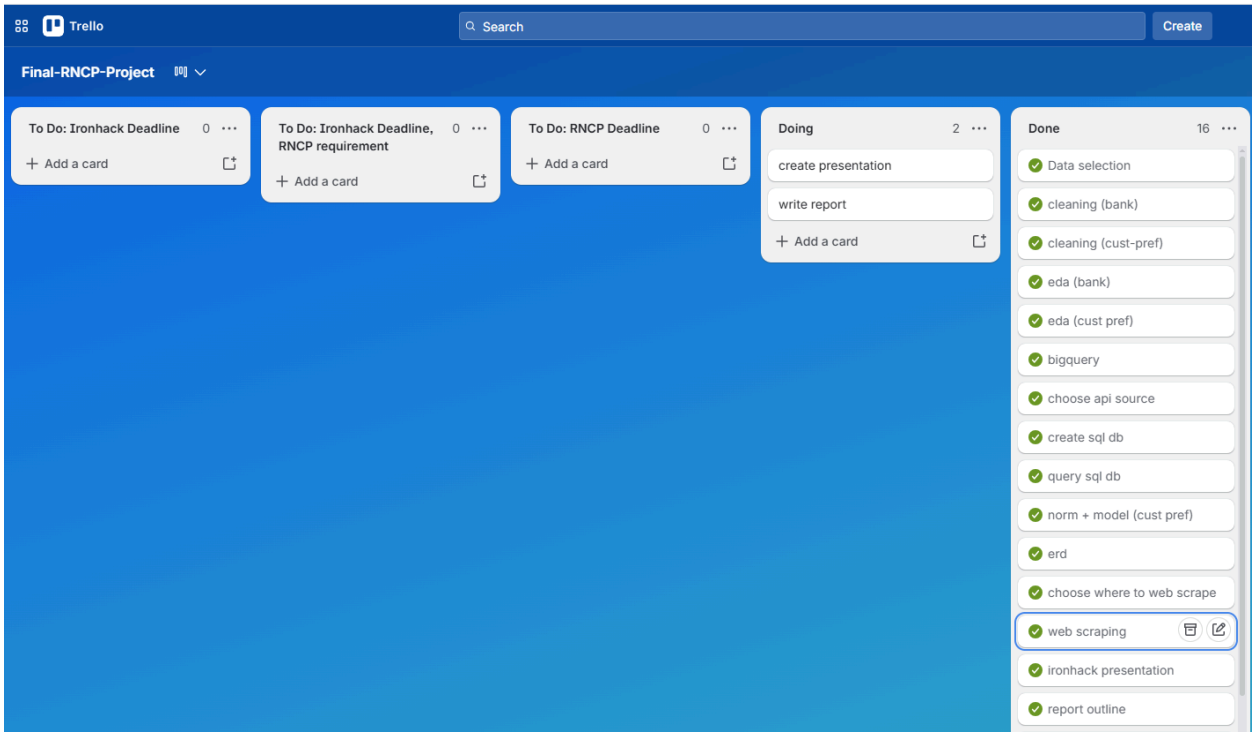
To enhance the UCI Bank Marketing dataset, additional socioeconomic information was incorporated from external sources.

Statistics Portugal (INE) provided official pension statistics used to estimate pension income for retired customers.
Eurostat supplied Portuguese minimum wage data through its public API.
Paylab.pt provided average salary information for occupations represented in the Bank Marketing dataset.

These external sources were used exclusively to enrich the Bank Marketing dataset by improving the representation of customer income and employment characteristics prior to modelling.

Project Management

The software Trello was used for project management:



Data collection

Data was collected from multiple source types including flat files, web APIs, and web-scraped data using automated Python scripts.

Flat File Import

The Customer Personality Analysis dataset was downloaded from Kaggle as a CSV file and imported using the Pandas library. As the primary project dataset, it provided the customer records used for segmentation and predictive modelling.

API Data Extraction

Several datasets were retrieved using APIs requests.

The UCI Bank Marketing dataset was accessed through the Hugging Face Datasets API.
Pension statistics were retrieved from the Statistics Portugal (INE) API.
Portuguese minimum wage data were collected from the Eurostat API.

Web Scraping

Occupation salary data were collected from Paylab.pt using a Python web scraper implemented with the BeautifulSoup library. The scraper extracted salary estimates corresponding to occupations contained within the Bank Marketing dataset, providing an additional socioeconomic feature for data enrichment.

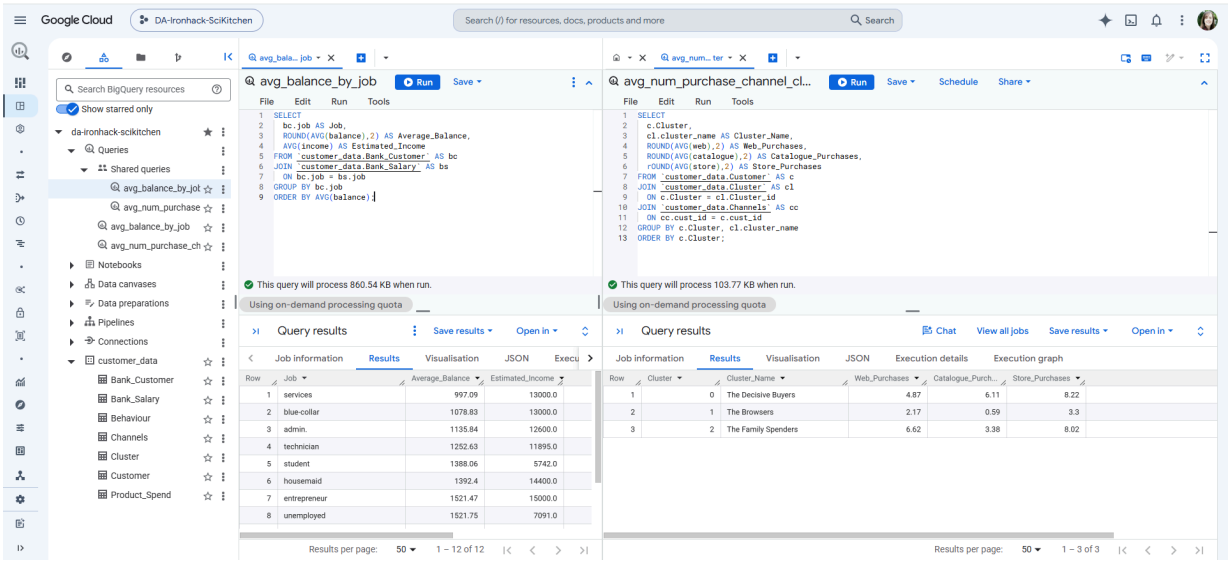
Data Integration

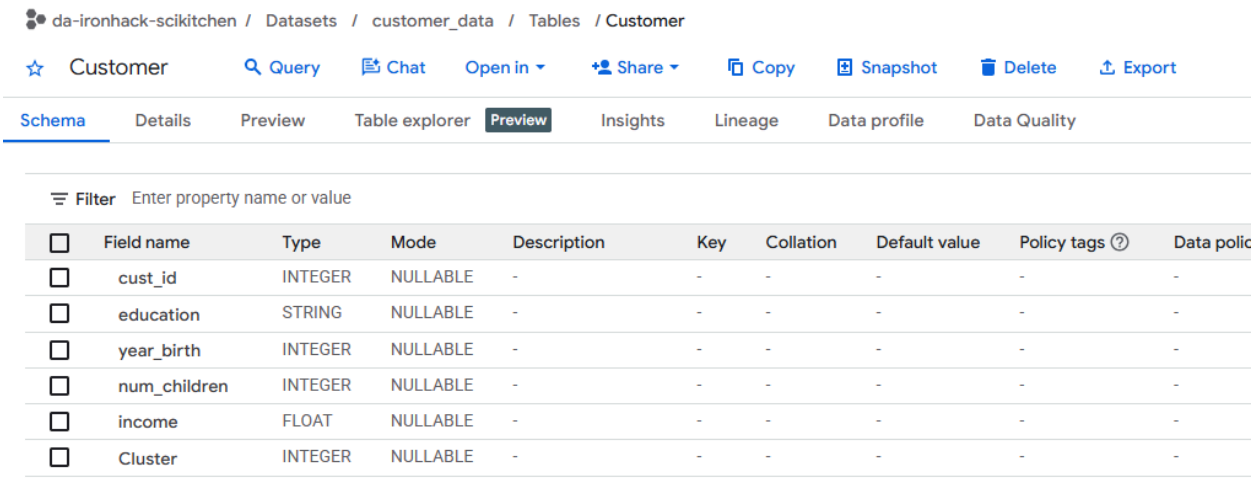
The external enrichment data (salary estimates, pension values, and minimum wage information) were matched to records within the UCI Bank Marketing dataset to produce an enriched dataset for comparative modelling. The Sci-Kitchen dataset remained unchanged and served as the primary dataset for segmentation and predictive model development.

Big Data Source: BigQuery

To demonstrate querying within a big data environment, the project used Google BigQuery as a cloud-based analytical database platform. As no suitable public BigQuery dataset was identified for this project, the project datasets were uploaded manually as seven CSV files and imported into BigQuery tables.Partitioning was not applied, as the largest table (bank_customer, ~40,000 rows) is small enough that scan cost and query speed aren't meaningfully affected either way. Clustering was considered but skipped for the same reason.

The uploaded tables represented the prepared project data and supporting datasets, allowing SQL-based extraction and analysis using BigQuery's query engine.





avg_num... ter

avg_num_purchase_channel_cl...

Run Save Schedule Share

File Edit Run Tools

```sql
SELECT
  c.Cluster,
  cl.cluster_name AS Cluster_Name,
  ROUND(AVG(web),2) AS Web_Purchases,
  ROUND(AVG(catalogue),2) AS Catalogue_Purchases,
  rOUND(AVG(store),2) AS Store_Purchases
FROM `customer_data.Customer` AS c
JOIN `customer_data.Cluster` AS cl
  ON c.Cluster = cl.Cluster_id
JOIN `customer_data.Channels` AS cc
  ON cc.cust_id = c.cust_id
GROUP BY c.Cluster, cl.cluster_name
ORDER BY c.Cluster;
```

This query will process 103.77 KB when run.

Using on-demand processing quota

Query results

Chat View all jobs Save results

Job information Results Visualisation JSON Execution details Execution graph

Row	Cluster	Cluster_Name	Web_Purchases	Catalogue_Purch...	Store_Purchases
1	0	The Decisive Buyers	4.87	6.11	8.22
2	1	The Browsers	2.17	0.59	3.3
3	2	The Family Spenders	6.62	3.38	8.02

Data cleaning and Exploratory data analysis

The primary dataset was already in a relatively clean state and required minimal preparation.

Handling of nulls: Null values were only found in the income column, these rows were removed as this was a core demographic feature, and there was insufficient other information to infer a dummy value.
Implausible values: Customers with ages over 100 (114, 115 and 121) were removed as they did not seem to reflect a simple input error. There was also a large gap between these values and the oldest age deemed plausible (74) providing further evidence of implausibility.
Values of “YOLO” and “Absurd” in marital status were replaced by “single”.
Consolidation of values: Marital status was consolidated to single, partnered (to include married and unmarried), and separated (to include divorced and widowed). Similarly, previous campaign responses were summed to give a total number of prior conversions, and number of kids and teens were consolidated to give a number of children.
Renaming of columns: Column names were standardised using lowercase letters and underscores to simplify recall.
From an initial dataset of 2,240 rows, data cleaning resulted in a final dataset with 2,212 rows.

Exploratory Data Analysis

The final dataset contained 2,212 customer records and 24 features. Approximately 15% of customers responded positively to the previous marketing campaign, resulting in a moderately imbalanced classification problem. The dataset contained five categorical variables (education, marital status, has children, previously complained, and conversion outcome) and 19 numerical variables.

Feature Correlations

A correlation matrix was generated for numerical variables to identify relationships between customer characteristics and potential redundancy between features. Strong correlations were observed between income and total spending (0.79), with income also strongly associated with spending across individual product categories (e.g., meat and wine spending: 0.69). Income showed a moderate relationship with previous campaign

success (0.37), suggesting that higher-income customers tended to demonstrate greater engagement.

The number of children showed a negative relationship with total spending, indicating that household composition may influence purchasing behaviour. These relationships suggest that income is an important driver of purchasing activity, while previous campaign engagement may provide a stronger signal for conversion prediction.

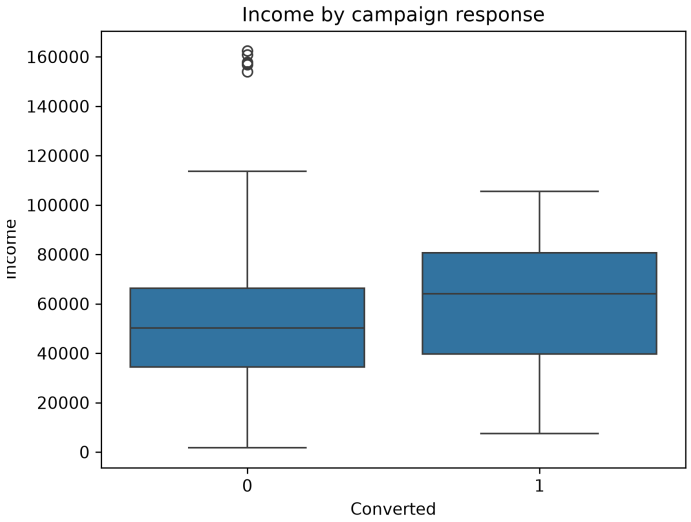
Relationship Between Features and Campaign Conversion

Exploratory analysis was conducted to identify variables associated with campaign response and guide feature selection and model development. Statistical tests and visualisations were used to evaluate relationships between customer characteristics and conversion outcome.

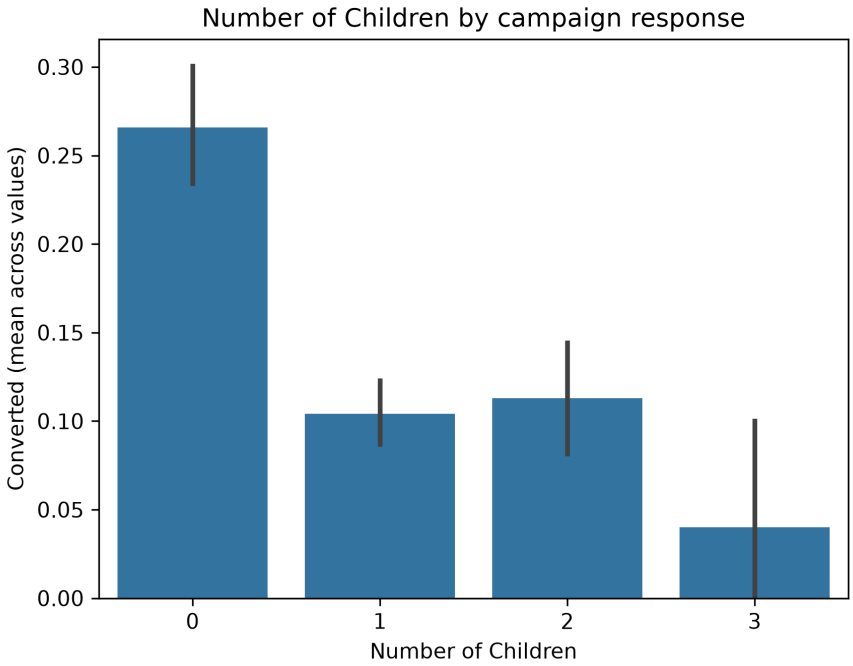
The strongest association with conversion was observed for previous campaign success. Customers with a greater number of successful responses to previous campaigns had substantially higher conversion rates (Welch's t-test, $p < 0.001$).

Several demographic characteristics were also significantly associated with conversion:

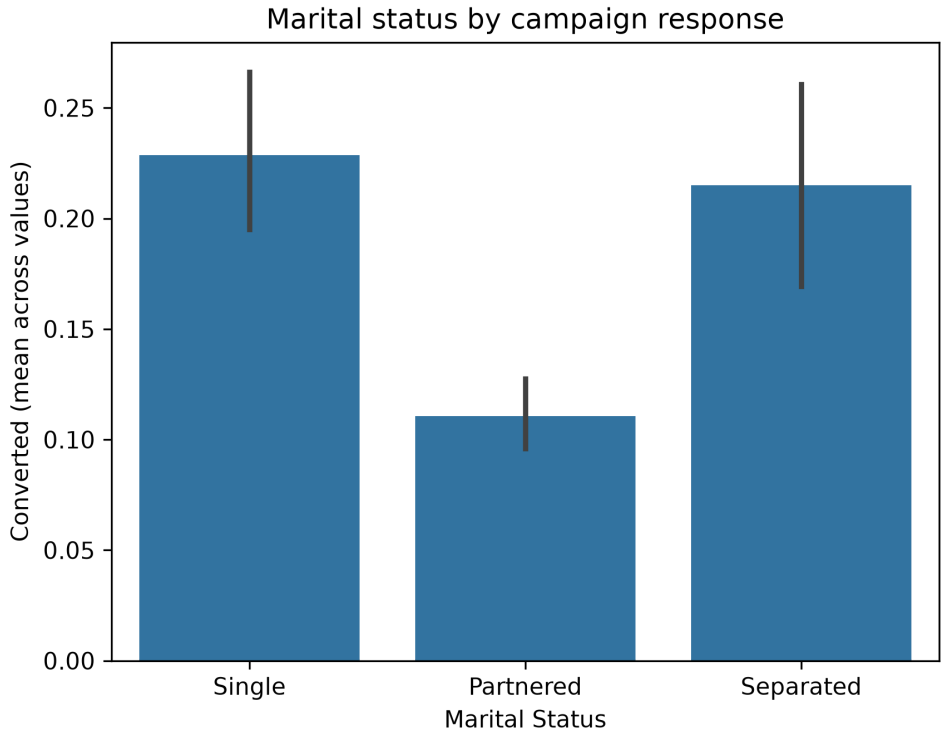
Customers with higher incomes were more likely to convert (Welch's t-test, $p < 0.001$). Higher education levels were associated with increased conversion probability (chi-square test, $p < 0.001$).



Household composition was significant, with conversion rates decreasing as the number of children increased (chi-square test, $p < 0.001$).



Marital status was associated with conversion, with customers living alone demonstrating higher conversion rates than partnered or separated customers (chi-square test, $p < 0.001$).



Behavioural variables also provided useful predictive signals. Total customer spending showed a significant association with conversion, suggesting that previous purchasing behaviour may help identify customers with higher likelihood of responding to campaigns.

Key Findings

Exploratory analysis identified previous campaign engagement as the strongest individual predictor of conversion, followed by demographic characteristics including income, education, and household composition. Behavioural indicators, particularly customer spending, also demonstrated predictive value.

These findings motivated the comparison of demographic, behavioural, relationship, and combined feature groups during model development to determine whether individual data sources were sufficient for accurate conversion prediction or whether combining multiple customer signals provided additional predictive power.

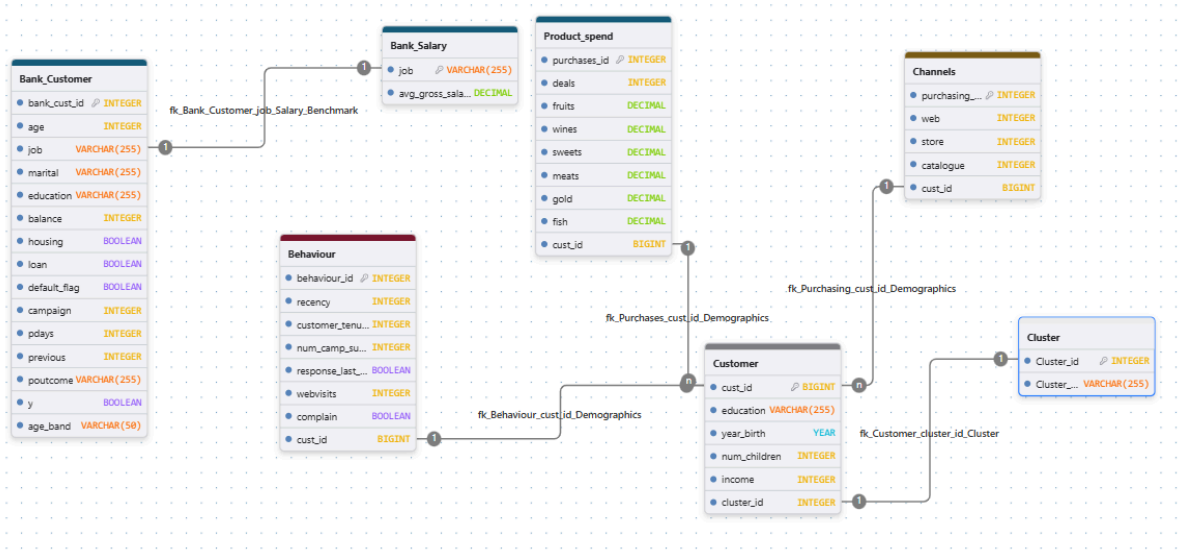
Database type selection

A relational database was selected for this project because the data consists of structured customer records with clearly defined relationships between customer demographics, behavioural metrics, purchasing activity, and campaign information. A relational model supports efficient storage of this structured data while enabling data extraction and aggregation through standard SQL.

MySQL was chosen as the database management system due to its widespread adoption and compatibility. It provides an accessible and reliable platform for storing customer data, enforcing data integrity through primary and foreign key constraints, and efficiently querying data.

Entities. ERD

The database consists of seven entities, each represented by a separate table. The schema contains five relationships, linking customer demographic, behavioural, purchasing, and segmentation data, together with the supplementary Bank Marketing and salary enrichment data. Primary and foreign key constraints were implemented to enforce referential integrity and minimise data redundancy. The complete logical structure of the database is shown in the Entity Relationship Diagram (ERD) below:



The database was then queried to obtain the following information:

```
USE ironhack_final_project;

SELECT *
FROM customer AS c
JOIN behaviour AS b
    ON c.cust_id=b.cust_id
JOIN channels AS ch
    ON c.cust_id=ch.cust_id
JOIN cluster AS cl
    ON c.cluster_id=cl.cluster_id
JOIN product_spend AS p
    ON c.cust_id=p.cust_id;
```

Result Grid

cust_id	education	year_birth	num_children	income	cluster_id	behaviour_id	recency	customer_tenure_in_days	num_camp_success	response_last_campaign	webvisits	complain	cust_id	purchasing_id	web	store	catalogue	cust_id	Cluster_id
1	Postgraduate	1957	0	58138	0	1	58	665	0	1	7	0	1	1	8	4	10	1	0
3	Postgraduate	1965	0	71613	0	3	26	314	0	0	4	0	3	3	8	10	2	3	0
12	Postgraduate	1959	0	63033	0	12	82	228	0	0	2	0	12	12	3	8	4	12	0
15	PhD	1946	0	82800	0	15	23	584	2	1	3	0	15	15	7	12	6	15	0
21	Postgraduate	1979	1	2447	0	21	42	541	0	0	1	0	21	21	0	0	28	21	0
28	PhD	1965	0	84618	0	28	96	221	1	0	2	0	28	28	6	10	9	28	0
33	Postgraduate	1946	0	68657	0	33	4	496	0	0	7	0	33	33	3	9	5	33	0
38	PhD	1943	0	48948	0	38	53	515	1	1	6	0	38	38	7	5	10	38	0
39	Postgraduate	1980	1	80011	0	39	3	428	0	0	4	0	39	39	8	5	6	39	0
43	Postgraduate	1976	0	79941	0	43	72	3	0	0	1	0	43	43	2	9	4	43	0
48	Postgraduate	1964	0	79143	0	48	2	689	0	0	3	0	48	48	6	13	9	48	0
50	Postgraduate	1977	0	82582	0	50	54	24	2	1	1	0	50	50	4	7	9	50	0

```
SELECT *
FROM bank_customer AS bc
JOIN bank_salary AS bs
    ON bc.job=bs.job;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

bank_cust_id	age	job	marital	education	balance	housing	loan	default	campaign	pdays	previous	poutcome	conversion	job	income
11	41	admin.	divorced	secondary	270	1	0	0	1	999	0	unknown	0	admin.	12600
12	29	admin.	single	secondary	390	1	0	0	1	999	0	unknown	0	admin.	12600
17	45	admin.	single	unknown	13	1	0	0	1	999	0	unknown	0	admin.	12600
26	44	admin.	married	secondary	-372	1	0	0	1	999	0	unknown	0	admin.	12600
33	60	admin.	married	secondary	39	1	1	0	1	999	0	unknown	0	admin.	12600
39	36	admin.	divorced	secondary	506	1	0	0	1	999	0	unknown	0	admin.	12600
40	37	admin.	single	secondary	0	1	0	0	1	999	0	unknown	0	admin.	12600
46	36	admin.	single	primary	-171	1	0	0	1	999	0	unknown	0	admin.	12600
54	42	admin.	single	secondary	-76	1	0	0	1	999	0	unknown	0	admin.	12600
61	32	admin.	married	tertiary	0	1	0	0	1	999	0	unknown	0	admin.	12600
81	45	admin.	single	secondary	206	1	0	0	1	999	0	unknown	0	admin.	12600
84	59	admin.	married	secondary	2343	1	0	0	1	999	0	unknown	1	admin.	12600
87	56	admin.	married	secondary	45	0	0	0	1	999	0	unknown	1	admin.	12600
93	60	admin.	married	secondary	290	1	0	0	1	999	0	unknown	0	admin.	12600
108	34	admin.	married	secondary	61	0	1	0	1	999	0	unknown	0	admin.	12600
117	41	admin.	married	secondary	351	1	0	0	1	999	0	unknown	0	admin.	12600
123	34	admin.	married	unknown	3	1	0	0	3	999	0	unknown	0	admin.	12600
127	29	admin.	single	secondary	818	1	1	0	1	999	0	unknown	0	admin.	12600
132	32	admin.	single	secondary	217	1	0	0	1	999	0	unknown	0	admin.	12600
134	55	admin.	single	secondary	802	1	1	0	2	999	0	unknown	0	admin.	12600
166	26	admin.	single	secondary	82	1	0	0	1	999	0	unknown	0	admin.	12600
169	54	admin.	married	tertiary	184	0	0	0	2	999	0	unknown	1	admin.	12600

```sql
SELECT
    bank_cust_id,
    balance,
    campaign,
    bc.job,
    income
FROM bank_customer AS bc
JOIN bank_salary AS bs
    ON bc.job=bs.job
WHERE balance <0
    AND campaign=1
ORDER BY balance ASC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

bank_cust_id	balance	campaign	job	income
15683	-6847	1	management	16800
1897	-3313	1	self-employed	15000
18574	-2827	1	technician	11895
31510	-2712	1	management	16800
25120	-2604	1	blue-collar	13000
20318	-1980	1	management	16800
35393	-1944	1	technician	11895
10625	-1941	1	housemaid	14400
15632	-1884	1	management	16800
11747	-1746	1	management	16800
5723	-1701	1	technician	11895
2365	-1664	1	blue-collar	13000

Result 17

```sql
SELECT
    cl.cluster_id,
    cl.cluster_name,
    count(cust_id) AS size_cluster
FROM cluster AS cl
JOIN customer AS c
    ON cl.cluster_id = c.cluster_id
GROUP BY cl.cluster_id
ORDER BY cl.cluster_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

cluster_id	cluster_name	size_cluster
0	The Decisive Buyers	510
1	The Browsers	1059
2	The Family Spenders	643

```
SELECT
    bc.job AS Job,
    count(bank_cust_id) AS Number_Customers,
    round(avg(income),2) AS Estimated_Income,
    round(avg(balance),2) AS Average_Balance
FROM bank_customer AS bc
JOIN bank_salary AS bs
    ON bc.job = bs.job
GROUP BY bc.job
ORDER BY count(bank_cust_id);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Job	Number_Customers	Estimated_Income	Average_Balance
unknown	288	11484.00	1772.36
student	938	5742.00	1388.06
housemaid	1240	14400.00	1392.40
unemployed	1303	7091.00	1521.75
entrepreneur	1487	15000.00	1521.47
self-employed	1579	15000.00	1647.97
retired	2264	7697.00	1984.22
services	4154	13000.00	997.09
admin.	5171	12600.00	1135.84
technician	7597	11895.00	1252.63
management	9458	16800.00	1763.62
blue-collar	9732	13000.00	1078.83

Data Dictionary

Customer Personality Analysis Schema

Cluster

Lookup table defining the three customer segments.

Field	Type	Description
cluster_id	INT	Unique segment identifier
cluster_name	VARCHAR	Segment name

Customer

Customer demographic information.

Field	Type	Description
cust_id	BIGINT	Generated customer identifier
education	VARCHAR	Highest education level
year_birth	YEAR	Year of birth
num_children	INT	Number of children in household
income	INT	Annual household income (€)
cluster_id	INT	Assigned customer segment

Behaviour

Customer relationship and campaign history.

Field	**Type**	**Description**
behaviour_id	INT	Behaviour record identifier
cust_id	BIGINT	Customer identifier

recency	INT	Days since last purchase
customer_tenure_in_days	INT	Customer tenure
num_camp_success	INT	Previous successful campaigns
response_last_campaign	BOOLEAN	Responded to previous campaign
webvisits	INT	Monthly website visits
complain	BOOLEAN	Customer complaint indicator

Product_spend

Customer spending by product category.

Field	Type	Description
purchases_id	INT	Spending record identifier
cust_id	BIGINT	Customer identifier
deals	INT	Purchases made using promotions
fruits	DECIMAL	Fruit spend
wines	DECIMAL	Wine spend
sweets	DECIMAL	Sweet product spend
meat	DECIMAL	Meat spend
gold	DECIMAL	Gold product spend
fish	DECIMAL	Fish spend

Channels

Purchasing channel usage.

Field	Type	Description

purchasing_id	INT	Channel record identifier
cust_id	BIGINT	Customer identifier
web	INT	Website purchases
store	INT	Store purchases
catalogue	INT	Catalogue purchases

UCI Bank Marketing Schema

Bank_Salary

Lookup table mapping occupations to estimated annual salary.

Field	Type	Description
job	VARCHAR	Occupation
income	DECIMAL	Estimated annual salary (€)

Bank_Customer

Customer demographics, financial information and campaign history.

Field	Type	Description
bank_cust_id	INT	Customer identifier
job	VARCHAR	Occupation
age	INT	Customer age
marital	VARCHAR	Marital status
education	VARCHAR	Education level
balance	INT	Average account balance
housing	BOOLEAN	Housing loan

loan	BOOLEAN	Personal loan
default	BOOLEAN	Credit default
campaign	INT	Contacts in current campaign
pdays	INT	Days since previous contact
previous	INT	Previous campaign contacts
poutcome	VARCHAR	Previous campaign outcome
conversion	BOOLEAN	Target variable (campaign conversion)

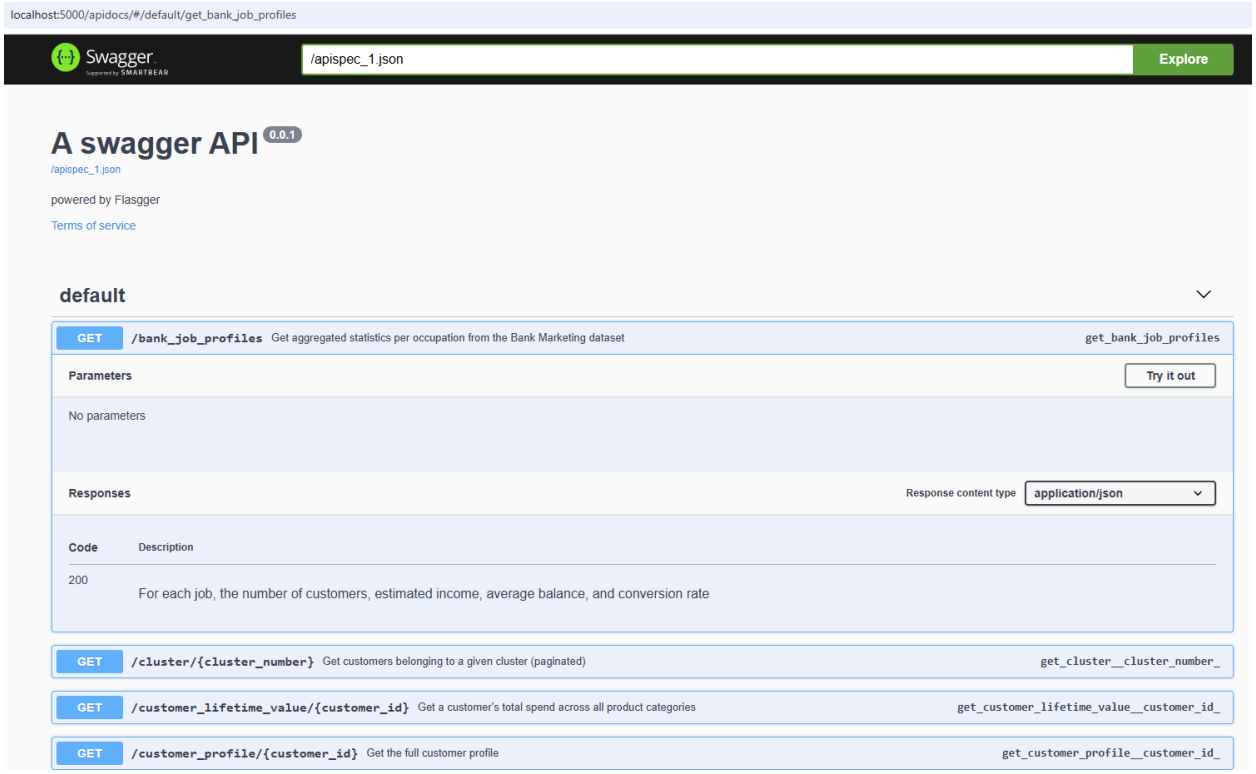
API

To enable access to the project data by external applications, a REST API was developed using Flask. The API connects to the MySQL database using the PyMySQL library and exposes selected customer and marketing data through HTTP GET endpoints. Database credentials are stored securely in a .env file, to avoid publicly sharing authentication details.

The API allows consumers to retrieve customer information, customer lifetime value, cluster membership, and aggregated Bank Marketing statistics. Each endpoint executes the required SQL query and returns the results as JSON, enabling integration with dashboards, reporting tools, or other notebooks.

The implemented endpoints are:

Endpoint	Description
/customer_profile/{customer_id}	Returns the complete customer profile by joining the customer, behaviour, purchasing, channel and cluster tables.
/customer_lifetime_value/{customer_id}	Calculates and returns the customer's total spend across all product categories.
/cluster/{cluster_number}	Returns all customers assigned to a specified customer segment.
/bank_job_profiles	Returns aggregated statistics for each occupation in the Bank Marketing dataset, including estimated income, average balance and conversion rate.



The /cluster/{cluster_number} endpoint supports pagination via page and page_size query parameters.

localhost:5000/cluster/1?page=1&page_size=10

Pretty-print ☑

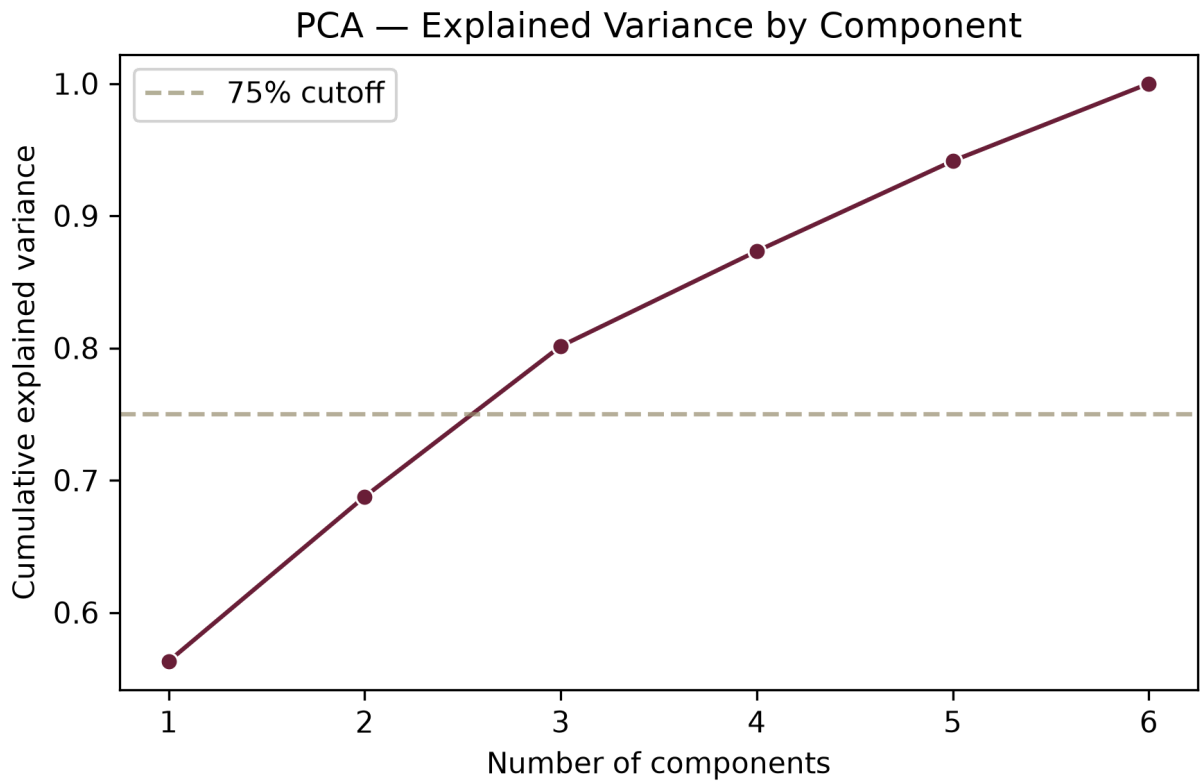

```
{
  "page": 1,
  "page_size": 10,
  "results": [
    {
      "cluster_name": "The Browsers",
      "cust_id": 2
    },
    {
      "cluster_name": "The Browsers",
      "cust_id": 4
    },
    {
      "cluster_name": "The Browsers",
      "cust_id": 8
    },
    {
      "cluster_name": "The Browsers",
      "cust_id": 9
    },
    {
      "cluster_name": "The Browsers",
      "cust_id": 10
    },
    {
      "cluster_name": "The Browsers",
      "cust_id": 11
    },
    {
      "cluster_name": "The Browsers",
      "cust_id": 14
    },
    {
      "cluster_name": "The Browsers",
      "cust_id": 16
    },
    {
      "cluster_name": "The Browsers",
      "cust_id": 17
    },
    {
      "cluster_name": "The Browsers",
      "cust_id": 19
    }
  ],
  "total_pages": 106,
  "total_results": 1059
}
```

Initial Findings & ML

Customer Segmentation

The six product spending variables demonstrated strong correlations, as customers who spent more tended to purchase across multiple categories. To reduce multicollinearity and create independent behavioural dimensions, Principal Component Analysis (PCA) was applied before clustering.

Three principal components were retained, explaining ~80% of total variance. These components were used as inputs for clustering, providing a reduced representation of customer purchasing behaviour while preserving the majority of the information contained within the original variables.



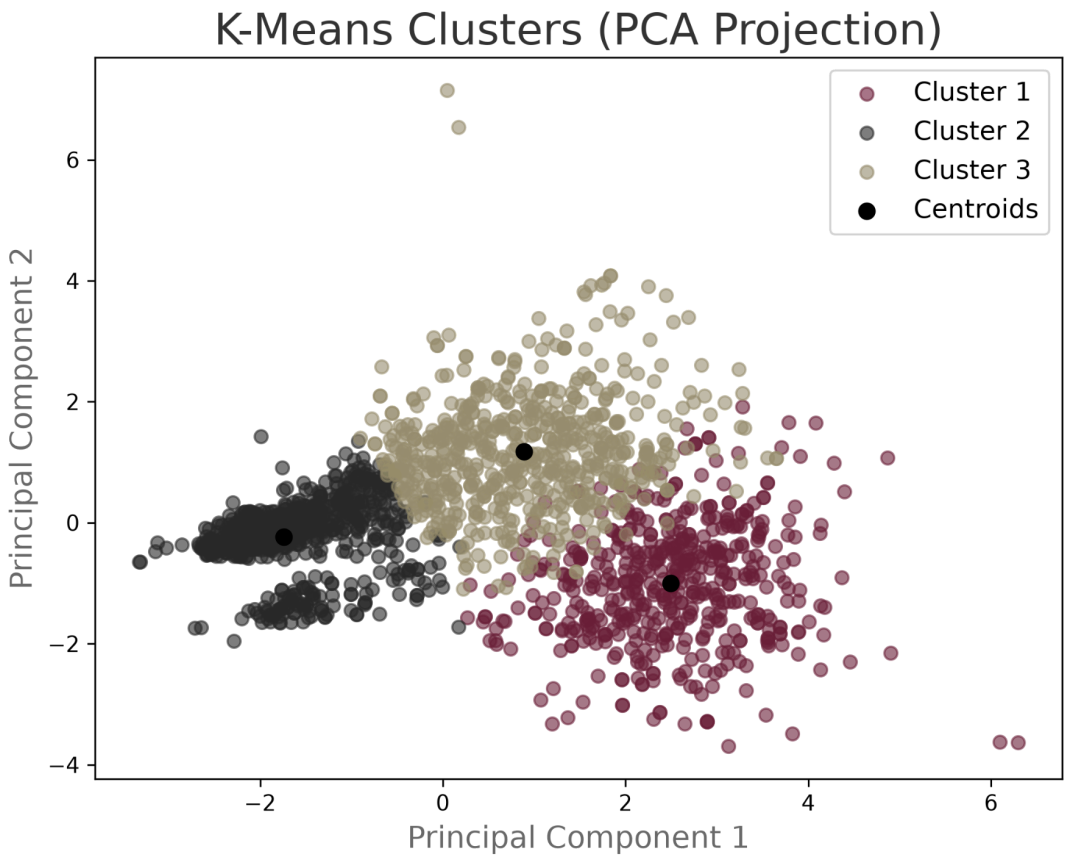
K-Means Customer Segmentation

K-Means clustering was applied to the PCA-transformed features. The optimal number of clusters was evaluated using silhouette score and inertia.

Both two and three clusters produced acceptable separation:

Two clusters: silhouette score = 0.3845
Three clusters: silhouette score = 0.3635

Three clusters were selected as the final solution because they provided greater marketing granularity while maintaining acceptable cluster separation.



Cluster profiles were then analysed using customer demographic, behavioural, and campaign features.

Segment	Size	Key Characteristics	Income	Avg Spend	Web Visits	Response Rate

1: Decisive Buyers	510 (23%)	High-income, high-spend customers with low digital engagement but highest conversion likelihood.	€76,437	€1,383	2.7	30.2%
2: Browsers	1,059 (48%)	High digital activity but low spending and conversion, suggesting lower purchase intent.	€34,699	€104	6.5	9.4%
3: Family Spenders	643 (29%)	Medium-high value customers who research purchases and respond to targeted campaigns.	€60,970	€822	5.5	12.3%

The clustering analysis identified three distinct customer segments with different commercial characteristics.

Segment 1 represents the highest-value customer group, characterised by high income, high spending, and the strongest conversion rate despite limited digital engagement.
Segment 2 represents the largest customer group but demonstrates lower purchasing value, suggesting that digital activity alone is not a reliable indicator of conversion likelihood.
Segment 3 represents a mid-to-high value segment that responds positively to targeted campaigns.

These differences highlight the importance of combining behavioural, demographic, and relationship signals when developing predictive marketing strategies.

Feature Set Ablation Study

To evaluate which customer signals contributed most to conversion prediction, models were trained using different feature groups:

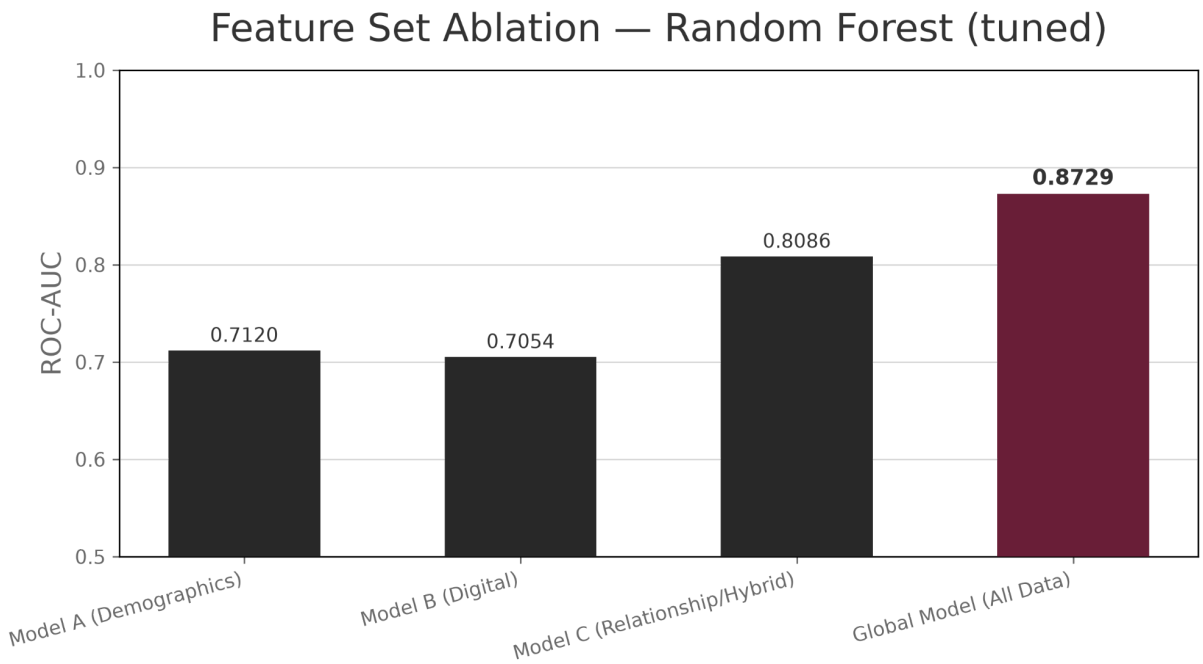
Demographic features
Behavioural/digital features
CRM/relationship features
Combined global feature set

A Random Forest classifier was initially selected for this comparison due to its ability to capture non-linear relationships and provide feature importance estimates.

Feature Set	ROC-AUC
Demographic	0.712
Digital Behavioural	0.705
Relationship / CRM	0.809
Global (All Features)	**0.873**

The results showed that the global model achieved the strongest predictive performance, suggesting that combining multiple customer signals provides additional predictive value compared with relying on individual feature groups.

The Relationship/ CRM model achieved the strongest performance among individual feature groups, highlighting the importance of previous customer engagement and relationship history.



Predictive Model Comparison

Final Model Selection and Validation

Following the feature-set ablation study, the global feature set containing all demographic, behavioural, and relationship variables was selected for final model optimisation. Three classification algorithms were evaluated: Logistic Regression, Random Forest, and XGBoost. Hyperparameter optimisation was performed using Grid Search. Models were compared using ROC-AUC.

Model	Accuracy	ROC-AUC
Logistic Regression	0.89	0.863
Random Forest	0.83	0.885
XGBoost	**0.90**	**0.902**

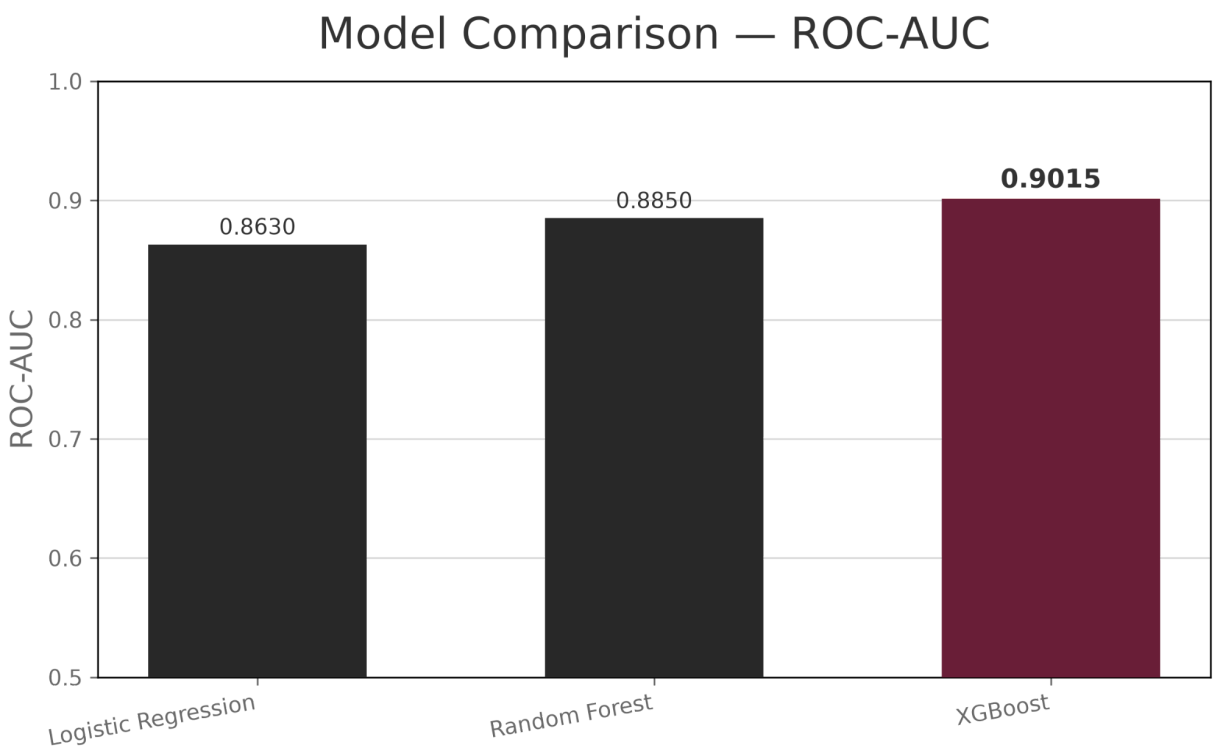
XGBoost achieved the highest predictive performance, with a ROC-AUC of 0.902, outperforming both Random Forest (0.885) and Logistic Regression (0.863). The final XGBoost model achieved a precision of 0.74 and recall of 0.45 for converted customers, demonstrating improved identification of likely responders while maintaining a low false positive rate.

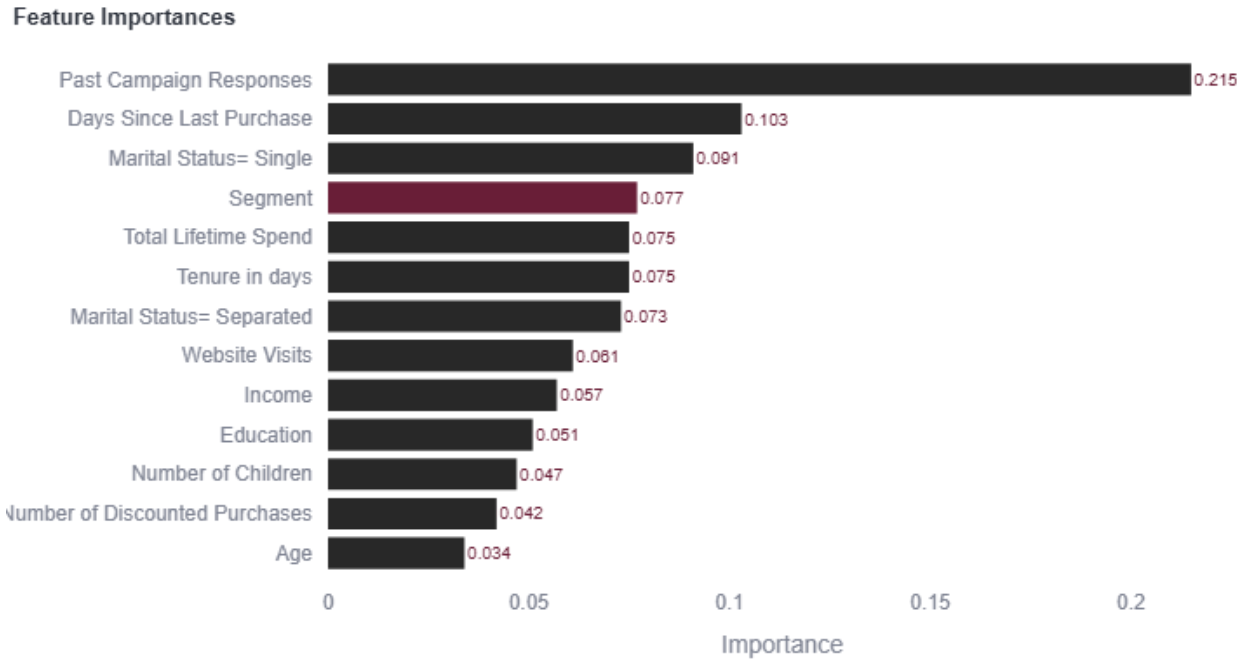
To determine whether the differences in ROC-AUC were statistically significant, DeLong tests were performed between model pairs.

Comparison	AUC Difference	p-value	Result
XGBoost vs Random Forest	+0.017	0.031	Significant
XGBoost vs Logistic Regression	+0.039	0.003	Significant
Random Forest vs Logistic Regression	+0.022	0.181	Not significant

The DeLong tests confirmed that XGBoost significantly outperformed both Random Forest and Logistic Regression. The difference between Random Forest and Logistic Regression was not statistically significant.

Based on its superior predictive performance and statistically significant improvement, **XGBoost was selected as the final predictive model for campaign conversion probability**.





External Validation: Customer Segmentation on UCI Bank Marketing

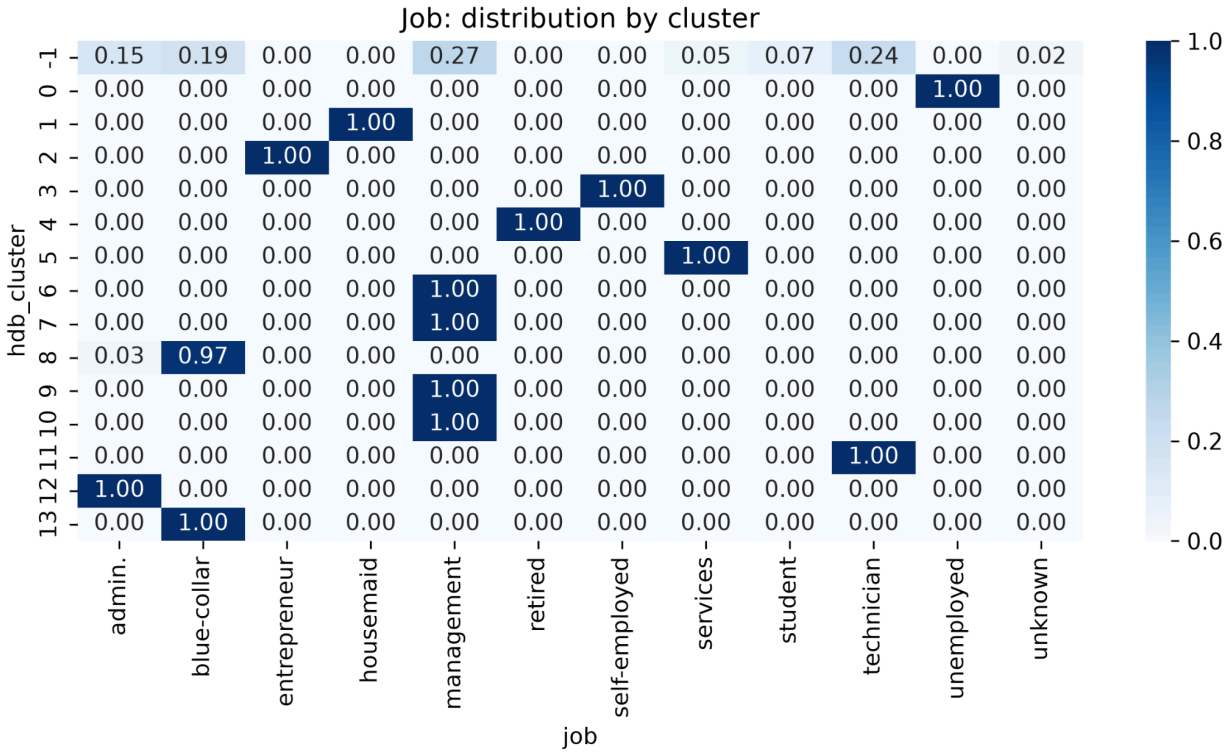
To evaluate whether the segmentation approach generalised beyond the Sci-Kitchen dataset, clustering was also applied to the UCI Bank Marketing dataset. Principal Component Analysis (PCA) was initially performed to reduce dimensionality; however, unlike the Sci-Kitchen dataset, the components did not reveal a strong low-dimensional behavioural structure. The first three components explained only 33.4% of variance, with 11 components required to capture approximately 76.7% of variance, suggesting that the dataset contained more distributed feature variation.

Density-based clustering using HDBSCAN was then evaluated. Different minimum cluster size parameters were tested; however, results showed increasing levels of noise and declining cluster quality as cluster size increased.

Minimum cluster size	Clusters identified	Noise (%)	Silhouette score
200	54	10.4%	0.543
500	30	23.9%	0.5422

1000	14	29.9%	0.445
2000	5	55.8%	0.472
3200	2	59.5%	0.413

Although smaller minimum cluster sizes produced acceptable silhouette scores, the resulting clusters were not meaningful. The identified clusters primarily represented occupation groups rather than distinct customer behaviour profiles. For example, several clusters consisted almost entirely of single job categories, including management, students, technicians, and retired customers.



Based on these observations, customer segmentation was not pursued further for the UCI dataset. The absence of strong behavioural segmentation was likely due to the dataset containing primarily demographic and campaign-contact variables rather than detailed purchasing or engagement behaviour.

Feature Ablation Validation

The feature-set ablation methodology was then repeated on the UCI Bank Marketing dataset using Random Forest classification.

Models were trained using demographic features, campaign-history features, and the complete feature set.

Feature Set	ROC-AUC
Demographic only	0.708
Campaign history only	0.762
Global (all features)	**0.780**

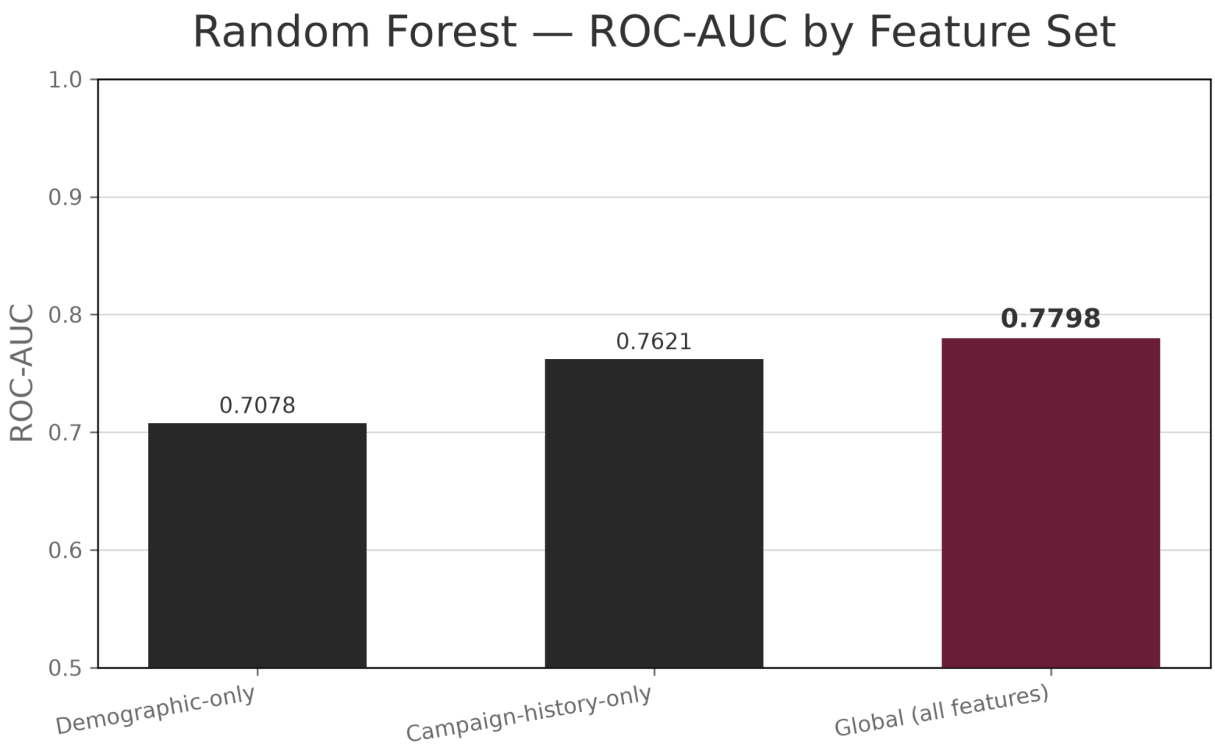
The results followed a similar pattern to the Sci-Kitchen analysis. Campaign-history features provided stronger predictive performance than demographic features alone, indicating that previous customer engagement and campaign interactions were more informative signals of conversion likelihood. The combined global model once again achieved the highest ROC-AUC.

DeLong tests were conducted to determine whether performance differences were statistically significant.

Comparison	AUC Difference	p-value	Result
Global vs Demographic	+0.072	<0.001	Significant
Global vs Campaign history	+0.018	<0.001	Significant
Campaign history vs Demographic	+0.054	<0.001	Significant

All pairwise comparisons were statistically significant, confirming that the observed differences in predictive performance were unlikely to be due to random variation.

These findings support the conclusion that customer relationship signals provide stronger conversion prediction capability than demographic characteristics alone, while combining multiple feature groups produces the most accurate models.



External Validation: Final Model Comparison

To assess whether the optimal modelling approach generalised across datasets, the final model comparison was repeated on the UCI Bank Marketing dataset using the global feature set. Logistic Regression, Random Forest, and XGBoost were evaluated using ROC-AUC as the primary metric.

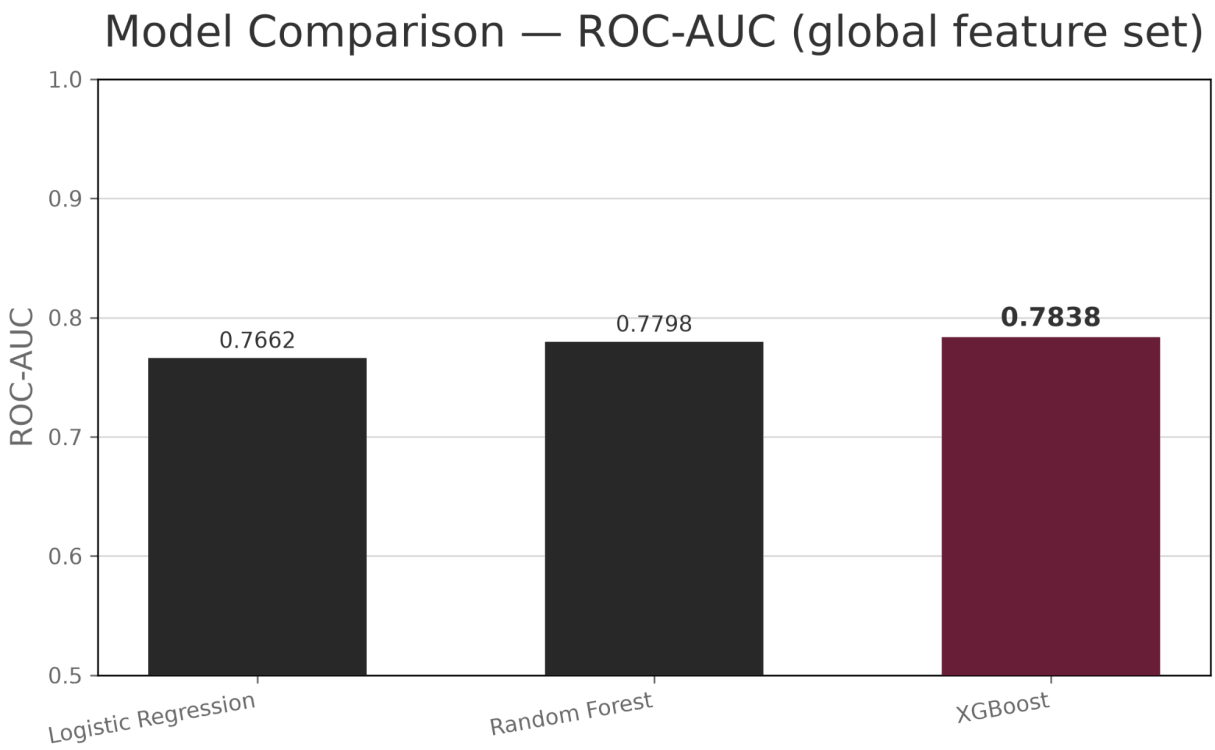
Model	Accuracy	ROC-AUC
Logistic Regression	0.76	0.766
Random Forest	0.81	0.780
XGBoost	0.89	**0.784**

XGBoost achieved the highest ROC-AUC (0.784), closely followed by Random Forest (0.780). Logistic Regression produced the lowest performance (0.766), indicating that tree-based models were better able to capture non-linear relationships within the customer data.

DeLong tests were performed to assess whether differences in ROC-AUC were statistically significant.

Comparison	AUC Difference	p-value	Result
Random Forest vs Logistic Regression	+0.014	<0.001	Significant
XGBoost vs Random Forest	+0.004	0.070	Not significant
XGBoost vs Logistic Regression	+0.018	<0.001	Significant

Although XGBoost achieved the highest ROC-AUC, its improvement over Random Forest was not statistically significant. Therefore, both models demonstrated comparable predictive performance, with XGBoost providing only a marginal improvement over Random Forest.



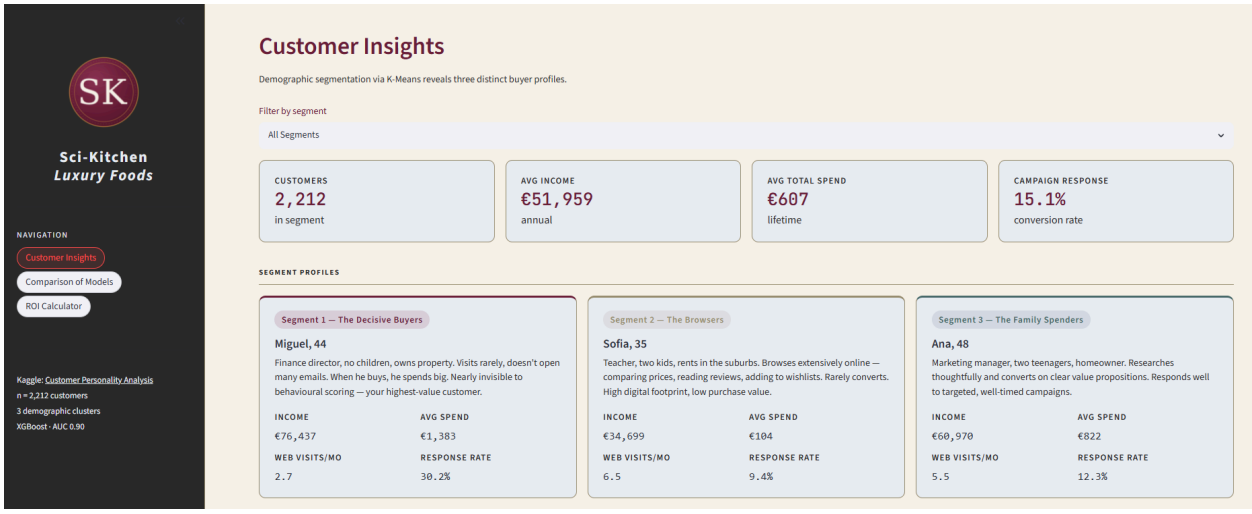
Streamlit Application Deployment

To demonstrate the practical application of the predictive segmentation and conversion models, a Streamlit application was developed to provide an interactive interface for exploring customer segments, evaluating model performance, and estimating campaign profitability.

The application contains three main pages:

Customer Insights:

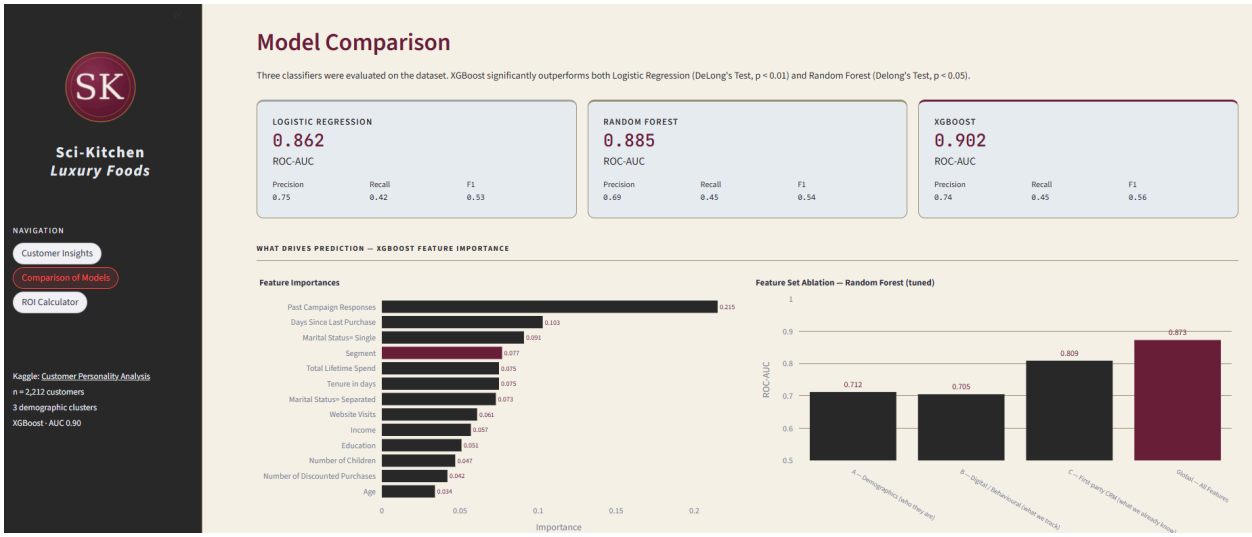
The segmentation page provides an overview of the three identified customer segments, including their demographic, behavioural, spending, and conversion characteristics. Users can select individual segments to analyse their profiles in isolation, allowing marketing teams to identify differences in customer value and engagement patterns.



Comparison of Models:

The model analysis page presents the performance of the evaluated classification models, including Logistic Regression, Random Forest, and XGBoost. Key performance indicators such as ROC-AUC, precision, recall, and accuracy are displayed alongside visual comparisons.

This allows users to understand both which model performs best and which customer signals contribute most to prediction accuracy. Feature-set ablation results are also shown.



ROI Calculator:

The final page provides an interactive campaign profitability calculator, allowing users to evaluate the commercial impact of different model choices.

Users can input:

- Campaign audience size
- Average product price
- Product margin
- Predicted conversion rate
- Cost per contact
- Selected Model

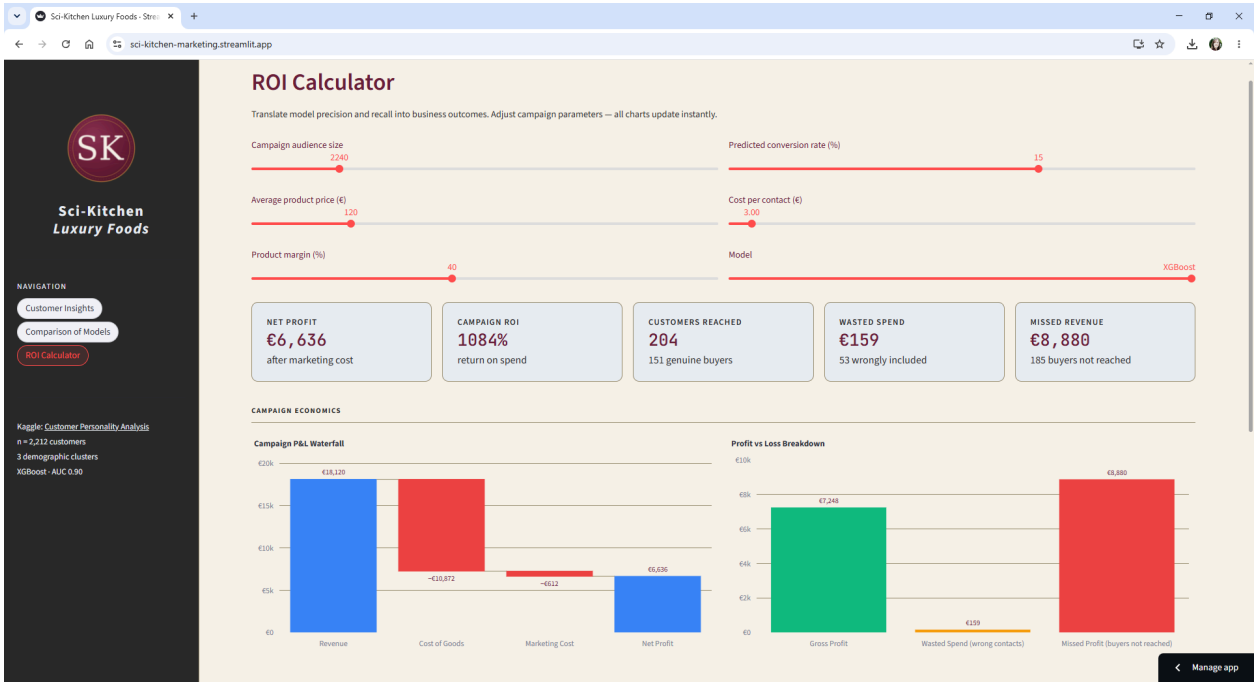
The simulator estimates campaign outcomes including:

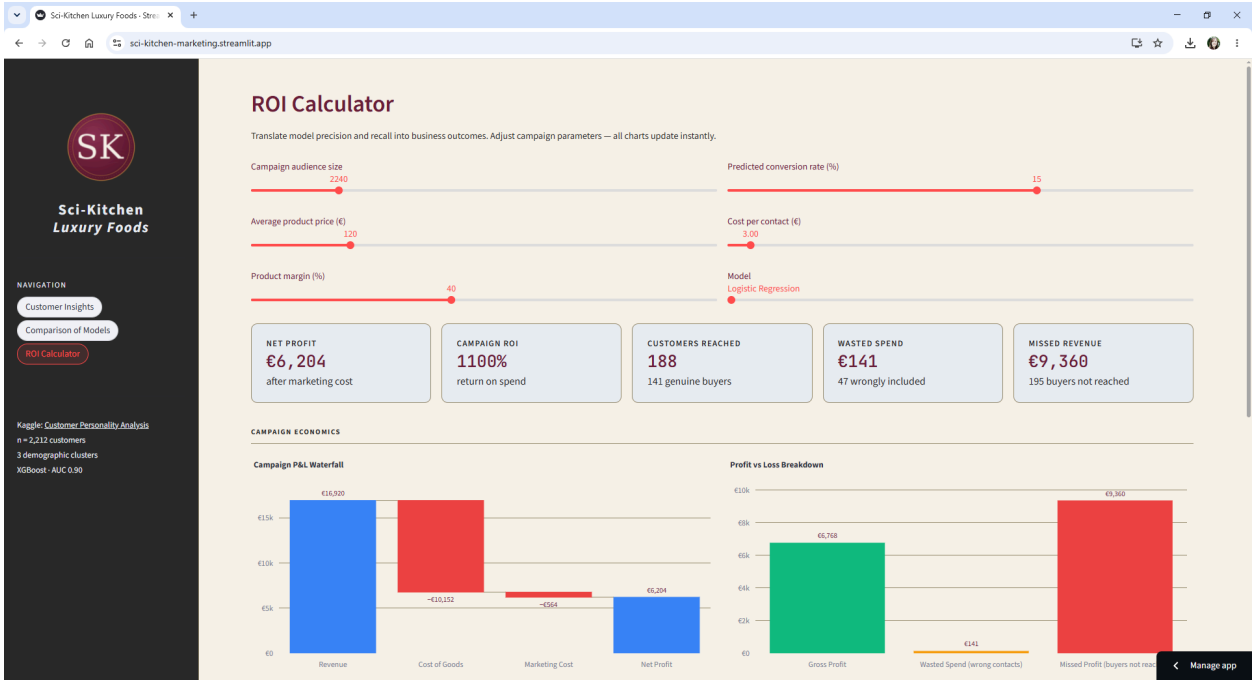
- Net profit
- Campaign ROI
- Customers reached
- Wasted spend
- Missed revenue

The ROI Calculator demonstrates that the optimal model depends on campaign characteristics rather than predictive performance alone. For example, XGBoost achieved the highest recall, making it suitable when missing potential customers is costly. However, Logistic Regression achieved higher precision, which may be preferable for campaigns with high contact costs where avoiding unnecessary customer outreach is more important.

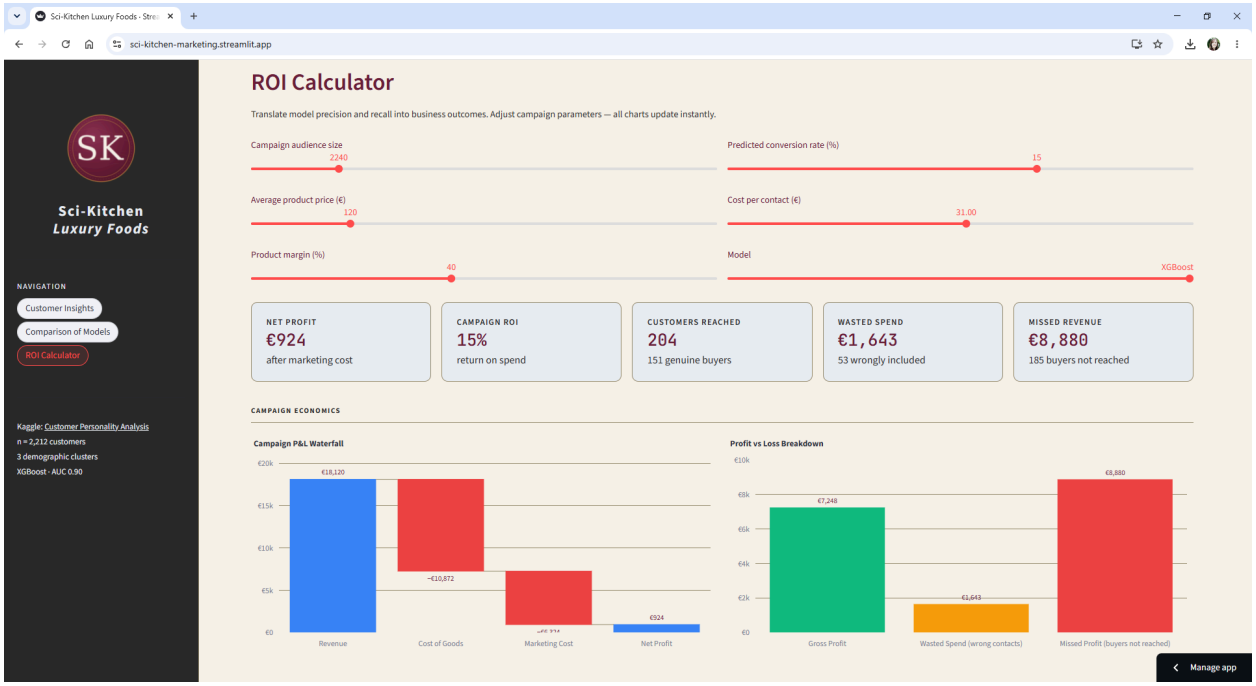
By linking model predictions to financial outcomes, the ROI calculator translates machine learning performance metrics into actionable marketing decisions.

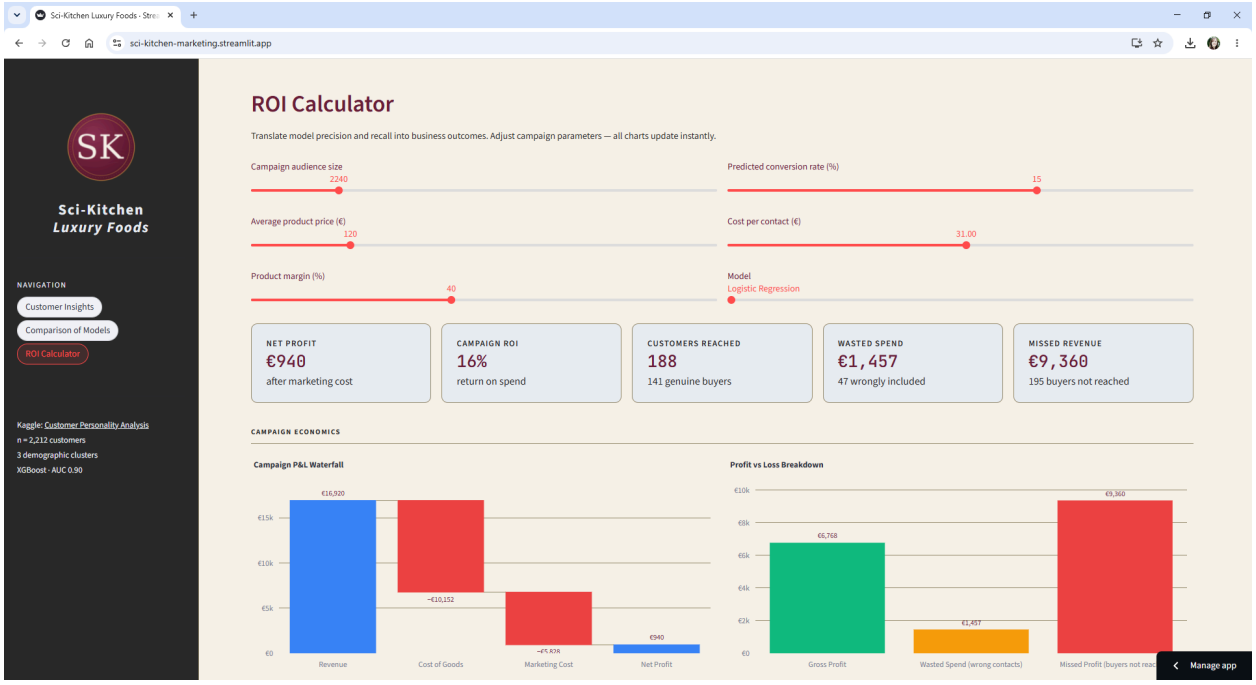
Case 1: For low campaign spend, a model with high recall is preferred as it minimises false negatives (i.e. missed revenue).





Case 2: For high campaign spend, a model with high precision is preferred as it minimises false positives (i.e. wasted spend).

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SK
Sci-Kitchen
Luxury Foods
NAVIGATION
Customer Insights
Comparison of Models
ROI Calculator
Kaggle: Customer Personality Analysis
n = 2,212 customers
3 demographic clusters
XGBoost · AUC 0.90
ROI Calculator
Translate model precision and recall into business outcomes. Adjust campaign parameters — all charts update instantly.
Campaign audience size
2240
Predicted conversion rate (%)
15
Average product price (€)
120
Cost per contact (€)
31.00
Product margin (%)
40
Model
Logistic Regression
NET PROFIT
€940
after marketing cost
CAMPAIGN ROI
16%
return on spend
CUSTOMERS REACHED
188
141 genuine buyers
WASTED SPEND
€1,457
47 wrongly included
MISSED REVENUE
€9,360
195 buyers not reached
CAMPAIGN ECONOMICS
Campaign P&L Waterfall
€16,920
−€10,152
−€5,828
€940
€15k
€10k
€5k
€0
Revenue
Cost of Goods
Marketing Cost
Net Profit
Profit vs Loss Breakdown
€10k
€8k
€6k
€4k
€2k
€0
€6,768
€1,457
€9,360
Gross Profit
Wasted Spend (wrong contacts)
Manage app

Overall Conclusion

This project aimed to determine which customer signals are most valuable for predicting marketing campaign conversion and whether these insights could generalise beyond the Sci-Kitchen retail context. By combining customer segmentation, feature-set ablation, and predictive modelling, the analysis demonstrated that conversion likelihood is best predicted through a combination of customer characteristics rather than relying on any single data source.

The Sci-Kitchen segmentation analysis identified three distinct customer segments based on purchasing behaviour. These segments demonstrated meaningful differences in spending patterns, digital engagement, and campaign response, highlighting that customers with similar purchasing behaviours respond differently to marketing approaches. Notably, the highest-value segment showed limited digital activity but the strongest conversion rate, demonstrating that online engagement alone is not a sufficient indicator of customer value or campaign likelihood. The segments identified also provide a framework for the development of audiences for outbound marketing campaigns, e.g. ad campaigns using Google Ads or LinkedIn.

The feature-set ablation study showed that relationship and campaign-history features provided the strongest predictive performance when used independently, outperforming demographic and digital behavioural features. This finding highlighted the importance of CRM hygiene and proper maintenance. However, the global model containing all available features achieved the highest ROC-AUC (0.873), confirming that demographic, behavioural, and relationship signals provide complementary information when combined. The final model comparison further improved performance, with XGBoost achieving the strongest Sci-Kitchen results (ROC-AUC 0.902) and statistically significant improvements over alternative models according to DeLong testing.

External validation using the UCI Bank Marketing dataset supported the broader findings. Although meaningful customer segmentation could not be replicated due to the more limited behavioural information available in the dataset, the predictive modelling results followed the same pattern. Campaign-history features outperformed demographic features, while the global feature set achieved the strongest performance (ROC-AUC 0.780). XGBoost achieved the highest ROC-AUC (0.784), although its improvement over Random Forest was not statistically significant.

The developed Streamlit application translated these analytical findings into a practical decision-support tool. Users can explore customer segments, compare model performance, analyse feature importance, and simulate campaign profitability. The ROI

calculator demonstrated that model selection should consider business objectives and the campaign itself rather than predictive metrics alone. It should be noted, however, that while initial results suggested that Logistic Regression achieved higher precision and XGBoost achieved higher recall, bootstrap analysis showed that the observed precision difference was not statistically robust and was likely attributable to sampling variation.

Overall, the results indicate that effective marketing prediction depends less on isolated demographic or digital activity indicators and more on understanding customer relationships, previous interactions, and combined behavioural signals. For Sci-Kitchen, the customer segments uncovered and final predictive model provide a practical approach for prioritising customers with higher conversion potential, reducing inefficient campaign targeting, and improving marketing return on investment.

GDPR

General Data Protection Regulation (GDPR) considerations were taken into account throughout the project. The datasets used are publicly available and do not contain directly identifiable personal data such as names, addresses, email addresses or telephone numbers.

For the Sci-Kitchen use case, customer identifiers were generated solely, and randomly, for database management purposes. Only the data required for analysis and modelling were stored and processed, in accordance with the GDPR principle of data minimisation.

The database and REST API expose only project data and do not provide access to personally identifiable information.

Relevant Links:

Kaggle dataset:
https://www.kaggle.com/datasets/imakash3011/customer-personality-analysis/data

Huggingface dataset: https://huggingface.co/datasets/cestwc/bank-marketing

EUROSTAT dataset:
https://ec.europa.eu/eurostat/databrowser/view/earn_mw_cur/default/table?lang=en

INE dataset:
https://www.ine.pt/bddXplorer/htdocs/minfo.jsp?var_cd=0014532&lingua=EN

Github repository: https://github.com/claireley/final-project

Trello dashboard:https://trello.com/b/VB9Asv4o/final-rncp-project

Streamlit application: https://sci-kitchen-marketing.streamlit.app/